

A counterexample to unitary lifting in operator-system ultraproducts

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2026-07-18

Source Problem

The source is Isaac Goldbring and Martino Lupini, *Model-theoretic aspects of the Gurarii operator system*, arXiv:1501.04332v3. In Section 5, after Question 5.5 on whether the class of operator systems unitally completely order isomorphic to a C*-algebra is elementary, they ask:

Question 5.6. With the preceding notation, if u is a unitary in $\prod_{\mathcal{U}} X_i$, are there unitaries $u_i \in X_i$ for which $u = (u_i)^{\bullet}$?

Here X_i are operator systems, $\mathcal{U}$ is an ultrafilter, and “unitary” is meant in the Blecher-Neal operator-space sense used in the source paper: $u \in X$ is a unitary if there is a linear complete isometry $X \rightarrow B(H)$ sending u to the identity operator.

Lemma 5.2. *Suppose that $\phi : A \rightarrow B$ is a ucp map between unital C^* -algebras that maps unitaries to unitaries. Then ϕ is a $*$ -homomorphism.*

We thank Thomas Sinclair for providing a proof for the following lemma.

Lemma 5.3. *Suppose that A is a unital C^* -algebra and $\dim(A) > 1$. Then there is a unital C^* -algebra B and a complete order embedding $\phi : A \rightarrow B$ that is not a $*$ -homomorphism.*

Proof. We first remark that A has a nonpure state. Indeed, since the states separate points and every state is a linear combination of pure states, we have that the pure states separate points. Since $\dim(A) > 1$, this implies that there are at least two pure states, whence any proper convex combination of these two pure states is nonpure.

Secondly, we remark that a nonpure state on A is not multiplicative. Indeed, if ϕ is a proper convex combination of the distinct pure states ϕ_1 and ϕ_2 , then taking a unitary u on which ϕ_1 and ϕ_2 differ, we have that $\phi(u)$ has modulus strictly smaller than 1.

We are now ready to prove the lemma. Suppose that A is concretely represented as a subalgebra of $B(H)$. Let ϕ be a non-pure state. Then the map

$$x \mapsto (\phi(x) \cdot 1) \oplus x : A \rightarrow B(H \oplus H)$$

is a complete order embedding that is not a $*$ -homomorphism. $\square$

Corollary 5.4. *No unital C^* -algebra is existentially closed as an operator system.*

Proof. This follows immediately from Lemmas 5.1, 5.2, and 5.3 (noting that existentially closed operator systems are infinite-dimensional). $\square$

Remark. Lemma 5.1 remains valid in the operator space category as well (with an identical proof). As a consequence, we see that if Z is an existentially closed operator space, then Z has no unitaries. Indeed, if Z is concretely represented as a subspace of $B(H)$, then the map

$$x \mapsto x \oplus 0 : Z \rightarrow B(H \oplus H)$$

is a complete isometric embedding into a C^* -algebra whose image contains no unitaries, whence, by Lemma 1, Z cannot contain any unitaries. In particular, we see that $\mathbb{N}\mathbb{G}$ contains no unitaries, a fact already observed (implicitly) in [27, Proposition 3.2].

Remark. Corollary 5.4 in particular shows that no unital exact C^* -algebra A is existentially closed as an operator system. We can be a bit more precise about how A fails to be existentially closed as an operator system. Indeed, since A is exact, by universality, there is a complete order embedding $A \hookrightarrow \mathbb{G}\mathbb{S}$. We claim that this embedding is not existential. Indeed, since $\mathbb{G}\mathbb{S}$ is existentially closed, if the above embedding were existential, then A would be existentially closed as an operator system, contradicting Corollary 5.4.

Given the above discussion, the following question seems natural:

Question 5.5. *Is the class of operator systems unitally completely order isomorphic to a C^* -algebra an elementary class?*

We now give a condition that would ensure a positive answer to Question 5.5. Suppose that $(X_i : i \in I)$ is a family of operator systems and $\mathcal{U}$ is an ultrafilter on I . If $u_i \in X_i$ is a unitary for each i , then it is clear that $(u_i)^\bullet \in \prod_{\mathcal{U}} X_i$ is a unitary of $\prod_{\mathcal{U}} X_i$.

Question 5.6. *With the preceding notation, if u is a unitary in $\prod_{\mathcal{U}} X_i$, are there unitaries $u_i \in X_i$ for which $u = (u_i)^\bullet$?*

We should note that the analog of Question 5.6 for C^* -algebras has a positive answer (see [12]).

Result

Question 5.6 has a negative answer. The counterexample is a sequence of three-dimensional commutative operator systems. Each factor contains no nonconstant unitaries, but the ultraproduct contains an element that is equal, in the ambient C^* -ultraproduct, to the usual coordinate unitary $z \in C(\mathbb{T})$.

Proof Intuition

The obstruction is non-stability of the metric equations defining unitaries in operator spaces. Put

$$f_\varepsilon(z) = z + \varepsilon z^2, \quad 0 < \varepsilon < 1/4.$$

Then f_ε is uniformly ε -close to the genuine unitary $z \in C(\mathbb{T})$. Thus if $\varepsilon_n \rightarrow 0$, the element $(f_{\varepsilon_n})^\bullet$ becomes exactly unitary in the ambient C^* -ultraproduct.

However, inside the small operator system

$$X_\varepsilon = \text{span}\{1, f_\varepsilon, \overline{f_\varepsilon}\} \subset C(\mathbb{T}),$$

there is no nearby unitary. The curve $z + \varepsilon z^2$ is strictly convex, so the scalar part of the Blecher-Neal norm test forces any operator-space unitary in X_ε to have modulus one at every point of $\mathbb{T}$. A Laurent-polynomial calculation then says that the only such elements in X_ε are scalar constants. These constants stay distance almost 2 from f_ε , so they cannot lift the ultraproduct unitary.

Lemma 1 (The factor unitaries are only constants). *Let $0 < \varepsilon < 1/4$, $f_\varepsilon(z) = z + \varepsilon z^2$, and*

$$X_\varepsilon = \text{span}\{1, f_\varepsilon, \overline{f_\varepsilon}\} \subset C(\mathbb{T})$$

with the inherited operator-space structure. Then the unitaries of X_ε are exactly the scalar constants of modulus one.

Proof. Let $v \in X_\varepsilon$ be a unitary in the operator-space sense. We first prove that $|v(\zeta)| = 1$ for every $\zeta \in \mathbb{T}$.

Consider the plane curve

$$\gamma_\varepsilon(\theta) = e^{i\theta} + \varepsilon e^{2i\theta}.$$

It is regular for $\varepsilon < 1/2$, and its signed curvature numerator is

$$1 + 6\varepsilon \cos \theta + 8\varepsilon^2.$$

It is also injective: if $z_1 + \varepsilon z_1^2 = z_2 + \varepsilon z_2^2$, then

$$(z_1 - z_2)(1 + \varepsilon(z_1 + z_2)) = 0,$$

and $|\varepsilon(z_1 + z_2)| \leq 2\varepsilon < 1$. Thus $z_1 = z_2$. For $0 < \varepsilon < 1/4$ the curvature numerator is strictly positive for all θ , so $\gamma_\varepsilon(\mathbb{T})$ is a strictly convex Jordan curve. In particular every point $\gamma_\varepsilon(\theta_0)$ is strongly exposed by some real affine functional on $\mathbb{C}$.

It follows that for every $\zeta_0 \in \mathbb{T}$ there is a self-adjoint element $h \in X_\varepsilon$ whose absolute value has a unique maximum at ζ_0 . Indeed, take an exposing functional $\ell(w) = \text{Re}(\eta w)$ for $w_0 = f_\varepsilon(\zeta_0)$, and replace $\ell(f_\varepsilon)$ by $R + \ell(f_\varepsilon)$ with R large enough that the maximum of the absolute value is the unique positive maximum.

By the Blecher-Neal characterization of unitaries, applied only at matrix level 1 and to the row $[v \ h]$,

$$\|[v \ h]\|^2 = 1 + \|h\|^2.$$

Since $X_\varepsilon \subset C(\mathbb{T})$, the left hand side is

$$\sup_{\zeta \in \mathbb{T}} (|v(\zeta)|^2 + |h(\zeta)|^2).$$

As $\|v\| = 1$ and $|h|$ has its unique maximum at ζ_0 , this equality forces $|v(\zeta_0)| = 1$. Since ζ_0 was arbitrary, $|v| = 1$ on $\mathbb{T}$.

Now write

$$v(z) = \alpha + \beta(z + \varepsilon z^2) + \gamma(z^{-1} + \varepsilon z^{-2}).$$

Multiplying by z^2 gives a polynomial

$$q(z) = \gamma\varepsilon + \gamma z + \alpha z^2 + \beta z^3 + \beta\varepsilon z^4$$

such that $|q(z)| = 1$ for every $|z| = 1$. A polynomial with constant modulus one on $\mathbb{T}$ is a scalar monomial cz^k . Comparing coefficients, the cases $k = 0, 1, 3, 4$ are impossible because $\varepsilon > 0$ links adjacent nonzero coefficients. The only possible case is $k = 2$, which gives $\beta = \gamma = 0$ and $|\alpha| = 1$. Thus v is a scalar constant of modulus one.

The converse is immediate: scalar constants of modulus one are unitaries in X_ε . $\square$

Theorem 1 (Negative answer to Question 5.6). *Let $\mathcal{U}$ be any free ultrafilter on $\mathbb{N}$. There are operator systems X_n and a unitary $u \in \prod_{\mathcal{U}} X_n$ that cannot be represented as $u = (u_n)^\bullet$ with each u_n a unitary of X_n .*

Proof. Set $\varepsilon_n = 1/(n+5)$, $f_n(z) = z + \varepsilon_n z^2$, and

$$X_n = \text{span}\{1, f_n, \overline{f_n}\} \subset C(\mathbb{T}).$$

Each X_n is a concrete unital self-adjoint subspace of $C(\mathbb{T})$, hence an operator system. Let

$$X = \prod_{\mathcal{U}} X_n, \quad A = \prod_{\mathcal{U}} C(\mathbb{T}),$$

and let $j : X \rightarrow A$ be the natural complete isometric inclusion. Put

$$u = (f_n)^\bullet \in X, \quad g = (z)^\bullet \in A.$$

Since $\|f_n - z\| = \varepsilon_n \rightarrow 0$, we have $j(u) = g$ in A . The element g is a unitary of the C^* -algebra A . Therefore

$$x \mapsto g^* j(x)$$

is a linear complete isometry from X into A sending u to 1_A . Thus u is a unitary of the operator space X , hence a unitary in the sense of Question 5.6.

Suppose, toward a contradiction, that $u = (u_n)^\bullet$ where each u_n is a unitary in X_n . By Lemma 1, $u_n = \lambda_n 1$ for some $\lambda_n \in \mathbb{T}$. For every $\lambda \in \mathbb{T}$,

$$\|f_n - \lambda\|_\infty \geq |f_n(-\lambda) - \lambda| = |-2\lambda + \varepsilon_n \lambda^2| \geq 2 - \varepsilon_n.$$

Consequently,

$$\|u - (u_n)^\bullet\| = \lim_{\mathcal{U}} \|f_n - \lambda_n\|_\infty \geq \lim_{\mathcal{U}} (2 - \varepsilon_n) = 2,$$

contradicting $u = (u_n)^\bullet$. Hence no such unitary lift exists. $\square$

Verification Notes

The accompanying script `code/verify_laurent_curve_checks.py` records the elementary scalar checks: the curvature lower bound $1 - 6\varepsilon + 8\varepsilon^2 > 0$ for the chosen range of ε , the monomial coefficient alternatives, and the distance bound from f_ε to scalar unitaries.

The proof is not a counterexample to Question 5.5 itself. It disproves the sufficient lifting condition proposed immediately after Question 5.5, namely Question 5.6.

Novelty and Literature Check

Before promotion, the cheap run indexes were searched for arXiv:1501.04332 and the phrases “Gurarij operator system”, “unitarydefinable”, “operator systems unitally completely order isomorphic”, and “ultraproduct unitaries”. No existing packet addressed Question 5.6.

External web search on 2026-07-18 used the exact label/text of Question 5.6 and close variants: “unitarydefinable operator systems ultraproduct unitaries”, “Is the class of operator systems unitally completely order isomorphic to a C^* -algebra”, and “unitaries operator systems ultraproduct Goldbring Lupini”. The search located the source arXiv page and the Blecher-Neal characterization paper, but no later paper explicitly answering Question 5.6.

References

References

- [1] I. Goldbring and M. Lupini, *Model-theoretic aspects of the Gurarij operator system*, arXiv:1501.04332v3, 2015. <https://arxiv.org/abs/1501.04332>
- [2] D. P. Blecher and M. Neal, *Metric characterizations of isometries and of unital operator spaces and systems*, Proceedings of the American Mathematical Society 139 (2011), 985–998; arXiv:0805.2166. <https://arxiv.org/abs/0805.2166>